QMS SOP TEMPLATE:

Environmental Monitoring Excursions

Quality
Management
System
Standard
Operating
Procedure

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Environmental Monitoring Excursions

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Introduction

Environmental Monitoring of areas where pharmaceutical product is manufactured, is essential for guaranteeing the microbial quality of the drug product.

Where aseptically or sterile produced products cannot be guaranteed for sterility by performing a sterility test on the product, Environmental Monitoring (EM) is the basis for warranting sterility.

Non-sterile products, are only required to have a bioburden as low as possible and be free of so called “objectionable microorganisms” and any other microorganisms that can harm the patient.

Table of Contents

1.0 Purpose

The purpose of this procedure is to describe the methodology used to manage Environmental Monitoring excursions/ data that are above alert or action limit or that show adverse data trends, in order to ensure data quality, regulatory compliance, and safety.

2.0 Scope

This procedure applies to all data related to Environmental Monitoring that are out-of-specification and/or out-of-trend.

3.0 Definitions/Abbreviations

Action Limit: A predefined threshold or value in a monitoring program that, when exceeded, requires a specific response, such as additional sampling, an investigation, or remediation.

Environmental Monitoring (EM): the systematic collection and analysis of data from the environment to assess its quality, detect changes, and ensure compliance with environmental regulations and standards.

QA/QC: Quality Assurance/Quality Control.

Out-of-Specification (OOS): any test results that do not meet the predetermined acceptance criteria, which are based on regulatory requirements, product quality standards, or manufacturing specifications.

Out-of-Trend (OOT): a result that differs significantly from historical data or expected patterns, suggesting a potential issue or anomaly, even if it’s not technically out-of-specification.

4.0 Responsibilities

- **QC Microbiology Manager (or designee):** Approves monitoring plans, ensures staff are trained, reviews trending reports.
- **QC Team Leader/Coordinator:** Leads the excursion investigation, administrates the process, ensures compliance with SOPs, organizes meetings on EM Excursions, summarizes information.
- **QC Laboratory Technicians:** Follow all instructions, operate equipment, collect and label samples properly.

5.0 Requirements

- Monitoring instruments (calibrated) or Electronic Monitoring System (qualified)
- Personal Protective Equipment (PPE)
- Sampling containers (sterile if necessary)
- Field data sheets/mobile data entry system
- Communication devices (radio/cell phone)

6.0 Risks/Points of attention

The Action Limits as defined in the GMPs are compliance limits, any excursion from those limits requires a thorough investigation.

However, plant specific Alert and Action limits are set on a statistical basis of previous Environmental Monitoring Data in similar conditions of operations, meaning that even when data show no excursion related to a GMP defined Action Limit, if an internal Action Limit is reached, a thorough investigation is required.

All different types of Environmental Monitoring Data (viable and non-viable particles, person, air or surface monitoring, air pressure differences, temperature and humidity control, bioburden or sterility testing, and microbial identification) form a basis of assessment and investigation for any excursion or abnormality since together they provide a full picture of the situation in which the excursion occurred.

7.0 Summary/Flowchart

7.1 Alert and Action limits are defined

7.2 Investigations when exceeding Alert and Action Limits

73 Trending of EM Data

7.4 Increase of Trending

8.0 Procedure

8.1 Alert and Action Limits

For all classified areas (A, B, C and D), alert and action levels should be defined with respect to the various viable and non-viable Environmental Monitoring data.

Action limits are in general set lower or at maximum equal to the specifications as defined in the GMP guidelines, depending on the variation of the data.

For grade A, alert limits are only on total particle level. For grade B, grade C and grade D alert and action limits should be set such that adverse trends are detected and can subsequently be addressed.

These alert and action limits shall be determined by using data from the qualification and validation studies and ongoing activities while in use. Statistical analysis of data shall define the most suitable alert and action limits for the respective areas and subareas.

8.2 When Exceeding Alert and Action Limits

If alert levels are exceeded, operating procedures should prescribe assessment and follow-up, which should include consideration of an investigation and/or corrective actions to avoid any further deterioration of the environment.

If action limits are exceeded, operating procedures should prescribe a root cause investigation, an assessment of the potential impact to product (including batches produced between the monitoring and reporting) and requirements for corrective and preventive actions.

Investigations can include:

- Identification of micro-organism(s) found, and their potential source (this is a requirement for action limits),
- Adjacent areas and their state of control,
- Any deviations or repair/maintenance activities related to utilities,
- Qualification status of operators present
- Workload/activities taken place in certain areas
- Changes in e.g., cleaning agents, cleaning staff or cleaning schedules

8.3 Data Trending

Besides the set alert and action limits to react on, there is also trending of data that gives information on the current state of control.

This trending should include, but is not limited to:

- Increasing numbers of excursions from action limits or alert levels;
- Consecutive excursions from alert levels;
- Regular but isolated excursion from action limits that may have a common cause, (e.g., single excursions that always follow planned preventive maintenance);
- Changes in microbial flora type and numbers and predominance of specific organisms. Particular attention should be given to organisms recovered that may indicate: a loss of control, deterioration in cleanliness and/or organisms that may be difficult to control such as spore-forming microorganisms and molds or that pose a specific risk to patients.

8.4 Increase of Monitoring and Trending

Monitoring and subsequent trending shall be increased in case of:

- Environmental monitoring excursions or other deviations
- Changes implemented in e.g., the area, utilities, cleaning agents
- Interventions in the environment (e.g., construction work in adjacent areas)
- Increased workload and/or presence of personnel

9.0 Documentation of Results

All documentation related to an Environmental Monitoring Excursion Investigation is stored in one file, if possible, to facilitate review of the Investigation in a later phase.

10.0 References

- 21 CFR Part 210 – Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs; General
- 21 CFR Part 211 – Current Good Manufacturing Practice for Finished Pharmaceuticals
- EU GMP Guide – Part I: Chapter 1 (Pharmaceutical Quality System)
- Pharmaceutical Inspection Co-operation Scheme. PIC/S Guide to Good Manufacturing Practice for Medicinal Products: Part I (PE 009-17)
- U.S. Food and Drug Administration. (2012). *Bad bug book: Handbook of foodborne pathogenic microorganisms and natural toxins* (2nd ed.)
- TRS 986 – Annex 2: WHO Good Manufacturing Practices for Pharmaceutical Products: Main

11.0 Revision History

- Rev 01 – Initial release
- Rev 02 – [To be updated with future changes]

12.0 Addenda

12.1 Addendum I: Objectional Microorganisms as mentioned in USP <1111> Example

- *Staphylococcus aureus*
- *Escherichia coli*
- *Pseudomonas aeruginosa*
- *Candida albicans*
- Bile-tolerant gram-negative bacteria

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